

HW # 35

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CS-F

$$A = \begin{bmatrix} 2 & 2 & -2 \\ 2 & -1 & 4 \\ -2 & 4 & -1 \end{bmatrix}$$

$$|A - \lambda I| = 0$$

$$\begin{vmatrix} 2-\lambda & 2 & -2 \\ 2 & -1-\lambda & 4 \\ -2 & 4 & -1-\lambda \end{vmatrix} = 0$$

$$2-\lambda ((1+\lambda)^2 - 16) - 2(-2-2\lambda+8) - 2(8-2-2\lambda) = 0$$

$$2-\lambda (1+\lambda^2+2\lambda-16) - 2(-2\lambda+6) - 2(6-2\lambda) = 0$$

$$2-\lambda (\lambda^2+2\lambda-15) - 4(-2\lambda+6) = 0$$

$$2\lambda^2 + 4\lambda - 30 - \lambda^3 - 2\lambda^2 + 15\lambda + 8\lambda - 24 = 0$$

$$-\lambda^3 + 27\lambda - 54 = 0$$

$$\lambda^3 - 27\lambda + 54 = 0$$

$$\lambda = -6 \text{ or } \lambda = 3$$

$$\lambda = -6$$

$$\left[\begin{array}{ccc|c} 8 & 2 & -2 & 0 \\ 2 & 5 & 4 & 0 \\ -2 & 4 & 5 & 0 \end{array} \right] \sim \left[\begin{array}{ccc|c} 1 & 0 & -1/2 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

$$x = \begin{bmatrix} 1/2 \\ -1 \\ 1 \end{bmatrix} x_3$$

$$\lambda = 3$$

$$\left[\begin{array}{ccc|c} -1 & 2 & -2 & 0 \\ 2 & -4 & 4 & 0 \\ -2 & 4 & -4 & 0 \end{array} \right] \sim \left[\begin{array}{ccc|c} 1 & -2 & 2 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

$$x = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} x_2 + \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix} x_3$$

eigenvectors :

$$\begin{bmatrix} 1 \\ -2 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix}$$

Not orthogonal

By gram Schmidt's

$$\begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix} - \frac{(-4)}{5} \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix} - \begin{bmatrix} -8/5 \\ -4/5 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} \quad \begin{bmatrix} -2/5 \\ 4/5 \\ 1 \end{bmatrix} \quad \begin{bmatrix} 1 \\ -2 \\ 2 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 2 & -2/5 \\ -2 & 1 & 4/5 \\ 2 & 0 & 1 \end{bmatrix} \begin{bmatrix} -6 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix} \begin{bmatrix} 1 & 2 & -2/5 \\ -2 & 1 & 4/5 \\ 2 & 0 & 1 \end{bmatrix}$$